《日德兰》（《JUTLAND》）重制版规则书 V1

本项目为**玩家自制重制内容，非营利目的**，仅供游戏交流使用
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前言：

本重制项目的全部资料均来源于互联网论坛及公开出版物。制作者自知才疏学浅，获取资料的渠道十分有限，学识水平亦多有不及，实难对内容的学术严谨性与观点正确性作出任何百分百的保证。对引用资料的误读、理解偏差，或因资料阙如而产生的主观臆断，恐怕在所难免，甚至可能错漏百出。为此，诚惶诚恐地恳请诸位同好、前辈与后来者，能够对书中存在的谬误予以毫不留情的批评指正。相较于学识深厚、兵棋推演经验丰富的前辈，鄙人不过是一块微不足道的垫脚石，唯愿能为来者略尽绵薄之力，抛砖引玉。

若有冒犯或疏漏之处，尚祈海涵。

一. 尺寸

1mm = 30yds

项目内所有距离单位为码

算子尺寸：20mm * 12mm [1]

移动尺：1 格 = 1kn/10min [2]

移动尺转向半径：15mm = 450yds [3]

二. 组件

文件夹 Game Components Assets 内：

决算表

Jutland Gun Resolve.xlsx

Jutland Torpedo Resolve.xlsx

船表

Ship Table.xlsx

算子板

Ship Counters.ai

Ship Counters Add.ai

测距尺

GE Range Finder.ai

GB Range Finder.ai

移动尺

Battle Maneuver Gauge.ai

罗盘

CompassV2.ai

战术桌垫

Tactical Battle Mat.ai

附件：Ship Top View.zip [4]

三．整体流程

战略流程与原作相同

战术流程：

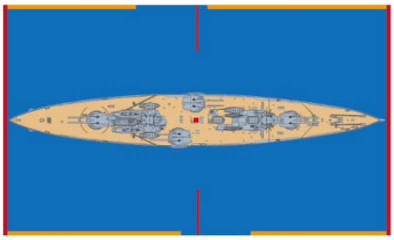
1. 投骰决定主动权，主动权方选择先后手（可选，旨在明确流程）
2. 发布命令（可选，见指挥通信节）
3. 先手宣布炮击目标，后手宣布是否规避机动
4. 后手宣布炮击目标，先手宣布是否规避机动
5. 先手半移动，后手半移动，放置鱼雷发射
6. 鱼雷命中判定
7. 后手全移动，先手全移动，放置鱼雷发射
8. 鱼雷命中判定
9. 炮击结算，鱼雷结算
10. 结算逐渐沉没、进水和修复状态

四．算子内容

船（编队）名

艉 —— 艏

Invincible



红色中心点：测距测角基准点

红色边垂线：移动基准点

边缘红色区：纵射射界

边缘黄色区：半齐射射界

边缘蓝色区：齐射射界

五．移动规则

用边垂线对齐移动尺进行移动&转向 [5]

航速即移动尺上对应前进格数，包括转向

半移动环节使用 航速/2（向上取整）移动

全移动环节使用 航速/2（向下取整）移动

若转向不足 45°，参考罗盘调整最终朝向（可选，配合指挥通信），按耗费 45°计算

轻型舰艇（轻巡编队、驱逐编队及单舰）转向不按移动尺，自由调整

单舰转向半径可以小于移动尺（可选）[3]

推荐编队航速 = 队内最低航速 - 2 [6]

六．船表内容 [8]

主力舰：[7]

<table>
<tr><th>Squadron</th><th>Ship</th><th colspan="7">Turret
Astern —— Bow</th><th colspan="10">Speed</th><th colspan="5">Armor</th><th colspan="8">Hull
1 slot = 5 HP</th><th colspan="5">Ammo storage
1 slot = 1/gun/min 1T</th></tr>
<tr><td rowspan="5">1BS
Flagship
5th Div.</td><td rowspan="5">**Colossus**</td><td colspan="7" rowspan="2">12"/50 Mark XI
5×2</td><td colspan="10">Sec: 8P 8S</td><td rowspan="2">0-10</td><td rowspan="2">10-20</td><td rowspan="2">20-30</td><td rowspan="2">30-40</td><td rowspan="2">TA</td><td></td><td></td><td></td><td></td><td>S</td><td></td><td></td><td></td><td colspan="5" rowspan="2">Status</td></tr>
<tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td></td><td>5</td><td></td><td></td><td></td><td></td><td></td><td>2</td></tr>
<tr><td>40</td><td>50</td><td></td><td></td><td>P</td><td>68</td><td>35</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td rowspan="3">6</td><td rowspan="3">7</td><td rowspan="3">5</td><td rowspan="3">5</td><td rowspan="3">D</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>100</td><td>100</td><td>100</td><td>100</td><td>100</td></tr>
<tr><td>Y</td><td>X</td><td></td><td></td><td></td><td></td><td>A</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td colspan="6" rowspan="2"></td><td>100</td><td>100</td><td>100</td><td>100</td><td>100</td></tr>
<tr><td>40</td><td>68</td><td></td><td>Q</td><td></td><td>50</td><td>35</td><td>21</td><td colspan="9"></td><td colspan="2"></td><td>100</td><td colspan="4"></td></tr>
</table>

Squadron：

编制，Flagship 则表明为旗舰

如图中 1BS Flagship 5th Div.表明为 1BS 旗舰，属于 5th Div.

若 1BS 6th Div. Flagship 则表明为 6th Div.旗舰

Ship：船名

Turret：炮塔及其命名 [9]

文字为火炮型号-炮座数-联装数

颜色 浅—深 表明炮塔甲板高度 低—高

若炮塔不相邻（如图中 A P），则表明有建筑遮挡

若炮塔相邻且有高度差(如图中 Y X),则表明是背负式炮塔(Superfiring Turrets)

若炮塔处于上/下行，则为侧舷炮塔

若炮塔朝向另一舷无遮挡，则可以跨舷开火

角落四个数字为向前、向后纵射射界（艏艉正向为 0）

靠内四个数字为跨舷齐射射界（艏艉正向为 0）

如图，可以得知 Colossus 左舷艏向纵射射界为 0-35，半齐射射界为 35-68，齐射射界为 68-90

如无数字默认为 30°

齐射角等相关规则见下一节

Sec：

副炮火力值，P = Port，S = Starboard

Speed：速度

德国船只航速存在争议，玩家可额外选用浅灰色区域航速使用 [10]

（如吕佐夫通常 26kn，可选 28kn）

Armor：

不同落弹角对应装甲值，TA：炮塔装甲值 [11]

Hull：

船体值，1 格 = 5 船体值（Hull Point）

2：若划去此格，航速 -2

5：若划去此格，航速 -5

S：若划去此格，进入逐渐下沉状态

Ammo storage：

弹药量，1 格 = 1 发/炮/分钟 在 10min 内的消耗 [12]

无论以多少炮塔开火，都按半数消耗（0.5 发/炮/分钟）或者全数（1 发/炮/分钟）划去单元格

状态： | ~~80~~ | ~~80~~ | 80 | 用尽 — 用半 — 未用

Status：用以记录状态

轻巡洋舰：[13]

Squadron	Ship	Attached to Fire power	Speed	Torpedo				Hull 1 slot = 5 HP	
4LCS-A	**Calliope** **Constance** **Caroline**	Independent T T T T 7 7 7 T T T T	Status 1 2 3 4 5 6 7 8 9 10 29	**21" Mark II**	Low **18kn** **18000**	Med **23kn** **15000**	High **45kn** **4500**	S 5 2 S 5 2 S 5 2	Status

Ship：编队内船只

Attached to：附属于（跟随）某个编队/船只

Fire Power：

相连的颜色相同区块表明一艘船的火力（顺序从右到左对应 Ship 列从上到下）

T = 1 个鱼雷管，上为左舷下为右舷

数字 X = 火力值

Torpedo：鱼雷性能表 [14]

第一列为型号

Low-Med-High 对应 低速-中速-高速 远-中-近 三个航程

Hull：

同上，每一行表明一艘船只（和 Ship 列从上到下对应）

装甲巡洋舰：

Squadron	Ship	Fire power	Speed	Torpedo & Armor					Hull 1 slot = 5 HP	
1CS	**Defence**	9.2"/50 Mark XI 2 × 2 T 15 T 10 10 T 15 T	Status 1 2 3 4 5 6 7 8 9 10 23	**18" Mark VII**	Low **19kn** **11500**	Med **28kn** **7000**	High **40kn** **3000**	All **3,G**	S 5 2	Status

Fire Power：

深色为主炮塔火力值，浅色为副炮火力值，鱼雷同上

齐射规则同主力舰

Armor：

All：落弹角 0-20° 的装甲值和炮塔防护值，20-40° 时此值 -1

驱逐舰：[15]

Squadron	Ship	Attached to Torpedo Tube 1=2	Speed	Torpedo				Hull + GunFire 1 slot = 5 HP	Torpedo Tube
1DF-A	**Fearless** **I class * 6**	5BS T T T T T 5 T T T	Fearless = 25 , Is =27 1 2 3 4 5 6 7 8 9 10 25 27	**21" Mark II**	Low **18kn** **18000**	Med **23kn** **15000**	High **45kn** **4500**	S 5 2 2 2 2 2 2 2	Status

Ship：独立命名为巡洋舰，“*数量”为驱逐舰

Torpedo Tube：

1T = 2 个鱼雷管，1t = 2 个待发鱼雷，单斜线表示取半

同上，相连的颜色相同区块表明一艘船的鱼雷，巡洋舰单列，巡洋舰格式同上

Speed：编队内不同船只航速

Hull+Gunfire：

巡洋舰为单独一行

驱逐舰为 1 格（驱逐领舰占 2 格），格内数字表明单艘驱逐舰火力值

七．炮击规则

齐射角规则：

背负式炮塔不能在纵射射界内同时开火 [16]

（如 Lion A B 炮塔在艏向 ±30° 开火内只算作 1 座炮塔）

仅 Tiger 的 Q 炮塔，Lützow Derfflinger 的 C 炮塔可以向后开火，且与艉炮塔在纵射射界内可以同时开火。其余中置炮塔或背负式炮塔不能向纵射射界开火。 [17]

侧舷炮塔可以向一侧的艏艉纵射射界开火，但不能跨舷纵射射界开火

（如 Colossus 的 P 炮塔可以向左舷任意角度开火，但不能向右舷开火除非齐射角）（而显然 Nassau 的 F 炮塔可以向左舷开火,但由于 E 炮塔阻挡不能在艉 20-37 半齐射射界内开火，也由于中央建筑阻挡不能向右舷开火）

回合内若跨舷开火，需结算 1D6 对自身损伤 [18]

算子齐射角使用办法：穿过目标中心点和己方中心点，穿过何色区即何种齐射角

数字测量的齐射角建议在算子模糊破损或者需要对照校验精确确定时使用 [19]

纵射射界：仅侧舷炮塔+非背负式炮塔可以开火，影响敌我炮击结算修正

齐射射界：全炮塔可以同时开火（六边形炮塔布局中被阻挡的另一侧炮塔除外）

半齐射射界：艏艉炮塔可以开火，中置炮塔不能开火

如上所述，此射界内单侧的侧舷炮塔可以开火，而对侧的侧舷炮塔不能开火

对于六边形炮塔布局，则视为向前射界时后部炮塔不能开火，向后同理

回合开始宣布目标后，须使目标一直处于某一射界内，若进入下一（火力更少）射界，则结算环节按火力更少的射界结算（建议此处因信称义，场上自由判断）

宣布炮击目标：

建议按照顺序分配目标而非逐个指定（关联后续指挥通信节，也符合史实）[20]

宣布射速 0.5 发/炮/分钟或 1 发/炮/分钟 [21]

规避机动：[22]

即通过驶向炮弹落点以干扰敌方校射的机动

原则上日德兰期间英方不使用此种机动，此处将其延伸为 Z 字机动

需先宣告，随后本回合航速按照-3kn 计算

炮击结算：

确定命中数：

1. 根据测距尺确定此距离基础命中率 [29]
2. 结算修正项
3. 根据齐射角状态确定开火的炮塔（管）
4. 射击数 = 炮管 * 射速 * 10min
5. 使用命中表计算命中数（若个位为 5 则取平均四舍五入计算，或直接用射数*命中率四舍五入，命中表就是这么来的，用于速查）
6. 根据命中数骰掷修正，得到最终命中数 [23]
7. 划去使用弹药格（使用 0.5 射速则划半，1 射速则划全）

例：Tiger 对 16500yds Lützow 射击，射速为 1，全程齐射角，炮塔无损
基础命中率 8%，缺乏训练（BCF）-1%，目标规避机动-3%，我舰被射击-1%
综合命中率 4%，查表得 3 命中数
3 对应 3-7 范围 2D6 - 7 ，骰掷 2D6 = 8 ，最终命中 3 + 8 - 7 = 4 发

查询损伤表：[24][25]

1. 查询火炮型号对于距离的落弹角
2. 查询目标落弹角下的装甲值（表或者测距尺）
3. 对于每一发命中骰掷 1D10
4. 查表，若
   (1) 1-2：则命中非装甲区
   (2) 3：则造成特殊损伤，查询特殊损伤表
   (3) 4-10：命中装甲区，用骰值+目标对应装甲值的结果索引对应列
5. 若索引格有数字，则扣除对应船体值
6. 若索引格有颜色，则查询对应效果表

例：Tiger 对 16500yds Lützow 射击，最终命中 4 发
此时 13.5"/45 Mark V(H)的 APC Mark Ia 落弹角为 10-20°
查船表，此处 Lützow 装甲值为 9
4D10 = 2，3，4，9
2 对应非装甲区，造成 2 点伤害并进水
3 对应特殊损伤
4+9=13，命中炮塔，查甲弹表
9+9=18，未击穿

甲弹表：

索引火炮、落弹角、目标 TA
根据字体颜色投骰判定击穿与否（无该 TA 则不可能击穿）
例：Tiger 对 16500yds Lützow 射击，命中 1 发炮塔
13.5"/45 Mark V(H) 10-20° 对 Lützow TA D 的字体为蓝色
1D6 = 5，未击穿

中口径结算：

对副炮、轻巡洋舰、装甲巡洋舰、驱逐舰火炮使用
轻巡洋舰和驱逐舰没有齐射角限制
巡洋舰射程 15000yds，驱逐舰射程 10000yds [26]
结算流程：

1. 计算开火总火力值
   (1) 集火限制（对单舰）见右列，超出集火限制则必须选择其他目标射击
   (2) 装甲巡洋舰、主力舰可以主副炮分别攻击两个目标
2. 先骰 1D6 得到乘数，乘以初始火力值，得到 FP 火力值（向下取整）
3. 根据“开火船只 - 命中船只”表得到修正火力值（向下取整）
4. 再根据距离表得到乘数，与修正火力值相乘得到最终伤害（向下取整）
5. 直接扣除船体对应伤害

八．伤害规则

驱逐舰编队：

每次对驱逐舰编队的伤害按照以下方式结算：

1. 计算总伤害量
2. 随机取编队中一艘驱逐舰，承受能承受的最大伤害（不大于总伤害量），总伤害量减去此次造成的伤害
3. 若总伤害量不为零，则重复以上过程
4. 若编队里存在巡洋舰，巡洋舰优先不承受驱逐舰命中的伤害，优先承受主炮伤害，对于其他命中的伤害，巡洋舰也参与上述流程，但每轮承受的最大伤害为 5

巡洋舰编队：

按照同样的轮转办法承受伤害

装甲巡洋舰：

按照主力舰方法通过炮击结算承受伤害

主炮（排除 AC）对巡洋舰、驱逐舰开火：

按照命中表计算，使用“射击轻型舰船”修正

进水：

带有此标识的舰船，每回合末（包括得到标识的初始回合）受到 1D10 伤害

承受伤害后骰掷 1D4，若结果<3（英）<4（德）即可移除（得到的回合可尝试移除）

3 回合后若此标识依旧存在，+1 进水标识，两个标识重新接着计算 3 回合

逐渐沉没：

航速-10kn

每 4 回合收到一个进水标识

九．鱼雷

每回合移动阶段可以自由放置鱼雷发射标识（使用算子替代）

发射方需记录发射的舰船、发射的数量、航速、航程

算子短边（12mm）用于 1-2 枚鱼雷的方向指示

算子长边（20mm）用于 3-4 枚鱼雷的方向指示

算子要准确放置朝向和发射点位置

放置后，对方玩家可立即宣布使用副炮或轻型舰船编队以发射编队为目标攻击

鱼雷放置后，下个半回合立刻开始向前移动，同时判定可能命中

如半移动阶段发射的鱼雷，阶段结束就可以判定，全移动阶段可以开始移动

全移动阶段发射的鱼雷，阶段结束就可以判定，下个回合的半移动阶段开始移动

移动直到抵达射程为止（因此需要两个标识）

此时判定的可能命中并不是真正命中

鱼雷命中判定：

以算子宽为宽度、鱼雷半移动距离（向上向下取整视阶段）为长度的区域称为鱼雷的危险区

若范围内有船只，不管敌我方，全部纳入判定

根据鱼雷发射数量，每枚鱼雷随机选取危险区内一艘船只作为目标

根据鱼雷航向与船只航向夹角，确定命中率 [27]

1D100 小于等于命中率则命中

撤走鱼雷标识

防御鱼雷攻击：

炮击结算阶段，对比攻击船只和发射编队的船只数量

若攻击船只数量大于发射编队船只，则鱼雷发射失败，撤回所有已发射（包括判定命中）的鱼雷

若攻击船只数量小于发射编队船只，两者相除得到比例，按比例随机撤回已发射（包括判定命中）的鱼雷（向下取整）

若攻击编队是由多艘驱逐舰组成，则其火力值可以按船只拆分，最终按攻击目标拆分计算数量比例

如：德国 4 艘、8 艘驱逐舰分别发起鱼雷攻击（并非 4、8 艘同时雷击，而是各自作为编队攻击）

英国 10 艘驱逐舰编队立刻宣布对两者发起攻击，拆分为 4 艘 和 6 艘驱逐舰

则最终结算，第一支德国编队的鱼雷全部撤回，第二支编队撤回 3/4 的鱼雷

若结算完毕有鱼雷成功发出，则船表上划去对应的鱼雷槽

鱼雷每阶段持续航行直至抵达射程

所有主力舰（BB、BC、B）两舷各有一点鱼雷值，且不能被防御 [28]

待发鱼雷：

每 2 回合可以装填 1 发待发鱼雷

鱼雷毁伤结算：

索引船只类型（区分 BB、BC 和其他船只），骰掷 1D10

结算方法同炮击结算

十．烟幕

烟幕的释放需要在移动前声明

烟幕的释放时机和鱼雷一样，在移动中的任一时刻均可释放

烟幕的长度位置跟随释放烟雾的船只的航行轨迹，可以通过算子定位连线的方式来表示

烟幕的释放只能通过驱逐舰或者巡洋舰

烟幕释放船只本回合航速-3kn

烟幕的持续时间为 3 回合

双方主力舰射击后的区域也会留下烟幕，持续时间为 1 回合

跨越两层烟雾的目标无法被射击

十一．指挥和通信

此部分仍需完善

不同船只一回合可以发送的信号数：

GB BF 旗舰	4 个
GB BCF 旗舰	3 个
GE BF 旗舰	3 个
GE 1SG 旗舰	3 个

信号类型	——	距离	——	特性
视觉信号	——	10000yds	——	穿过烟幕-25%
光学信号	——	28000yds	——	占 2 信号位
无线电	——	长	——	占 2 信号位（德 1.5），延迟 1 回合

信号内容（1 行 = 1 个）

准备/攻击	——	火力分配（A to B）（顺 X 序射击）
机动	——	航向+速度 / 前进 X 格+转向
编队	——	编组队形（展开）
问询	——	
回答/报告	——	
表达意图/职责	——	指派鱼雷攻击 etc
执行机动	——	

无线电仅驱逐领舰以上船只配备

英国舰队在未收到信号的情况下不会开火

引用和解释：

[1]参考《Battlelines and Fast Wings: Battlefleet Tactics in the Royal Navy, 1900－1914》Stephen McLaughlin

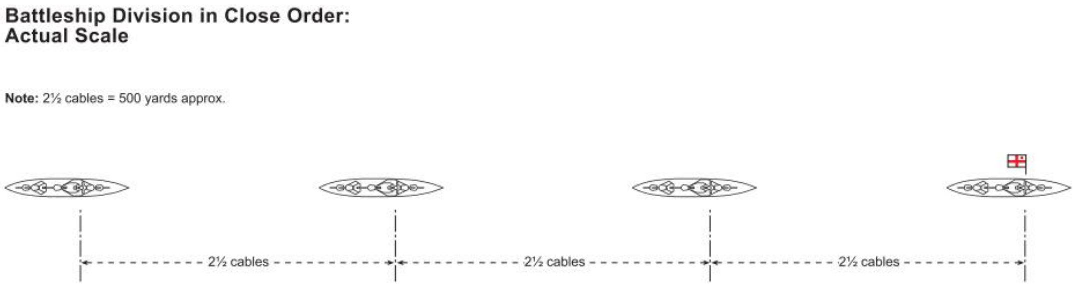


Figure 1. According to the Admiralty publication *Remarks on Handling Ships* (1911), the distance was to be measured as if it were from stem to stem, with allowance made for the position of the observer and the point used to measure the distance on the ship next ahead; sextants or rangefinders could be used to measure the distance to the mainmast of the ship ahead, after which the distance could be kept steady 'by selecting some temporary mark in your own ship to coincide with another in the Ship ahead, the relative position of which will indicate whether the ship is dropping or gaining'. (Diagram copyright John Jordan)

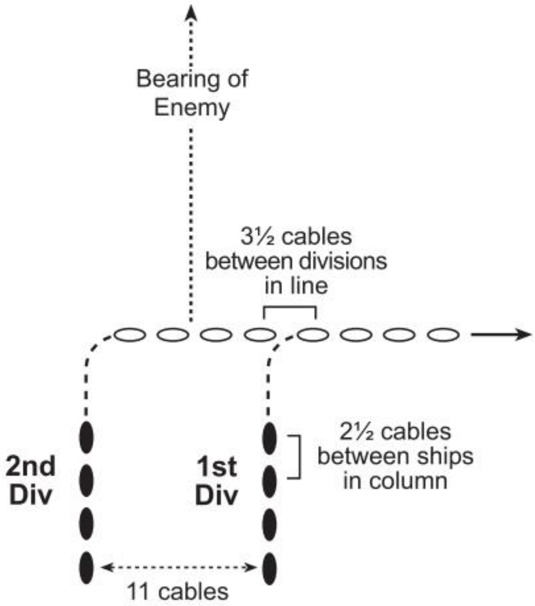


Figure 2. The divisions are shown at manoeuvring distance – the interval (2½ cables or 500 yards) times the number of ships in the division, plus one cable, as specified in Jellicoe's wartime Grand Fleet Battle Orders. (Diagram copyright John Jordan)

及 Tactical Orders and Doctrine, Battleships and Cruisers, U.S. Navy

B. TACTICS-1938-1942-In the U.S. Navy the basic evolution unit in battleship tactics is the two-three ship division(BATDIV). Most of the 1938-1942 battleships are organized in three ship divisions. Standard distance for individual ships is 700 yards, whth interval between divisions set at 2100 yards.

船只算子长度应当在500-700码以内才能保证准确的战列线间距模拟，否则需要大量的堆叠状态，这对尺规模拟是很麻烦且缺乏美感的。

[2]感谢《AH Jutland rules thoughts and modifications (v5)》Brian McCue 其中 Scale 部分的论述，对于原作的比例换算的阐释十分详尽。

文章来源：

https://boardgamegeek.com/filepage/132992/ah-jutland-rules-thoughts-and-modifications

[3]参考《British Battleships of World War One》R A Burt

***DREADNOUGHT*: TURNING TRIALS**

All held in English Channel

Run:	**1**	**2**	**3**	**4**	**5**	**6**
First 4 points turned in (seconds)	36	25	27½	32 7/10	34 4/5	39
Distance in turning (yards)	359	268	156	165	175	169
Speed (knots)	17.72	14.8	10.9	9.02	8.9	7.71

估计 180° （16 个罗经点）的转向时间在 1.5 - 3min 之间

***BELLEROPHON* CLASS: TURNING TRIALS, 1908–9**

Tactical diameter (at full speed):

***Bellerophon*:** 445 yards; rudder put over in 8 seconds

***Temeraire*:** 412 yards; rudder put over in 11¾ seconds

***Superb*:** 405 yards; rudder put over in 10 seconds

Comparison with earlier battleships:

***Agamemnon*:** 386 yards (17 knots)

***King Edward VII*:** 438 yards (18 knots)

***Duncan*:** 530 yards (18 knots)

***Dreadnought*:** 408 yards (18.12 knots).

***IRON DUKE*: STEAM TRIALS, 1932**

Displacement: 22,885 tons

Draught: 24ft 9¾in forward, 26ft 9⅛in aft

Wind: 1–2

Sea: Slight swell

Course: Measured mile at Tolland

First run	242rpm	14,563shp	17.062 knots
Second run	241.5rpm	14,469shp	18.127 knots
Third run	241.5rpm	14,548shp	16.925 knots
Fourth run	243rpm	14,749shp	18.237 knots

Tactical diameter: Helm put over at 15° which gave a circle of 700 yards, taking 5 minutes and 45 seconds to turn 180°

及 Tactical Orders and Doctrine, Battleships and Cruisers, U.S. Navy

Tactical diameter is generally 1000 yards. For zigzagging tactical diameter will be 2000 yards.

及《Handling a Battleship》Lieutenant Scott A. Robinson, U. S. Navy

https://www.usni.org/magazines/proceedings/1988/april/handling-battleship

"... The battleship will turn in a tight circle at speeds as low as 15 knots. At 15 knots with hard rudder (35°), a 180° turn can be completed with a tactical diameter of 760 yards in about two minutes. This tactical diameter does not increase appreciably with higher speed turns, although the time to come about does drop rapidly. ..."

考虑到铁公爵的转向时间是在小舵角的情况下,理论上全舵角的转向半径和时间应当更小，和上文的 2min 数据相近似。因此应当可以认为战列舰的转向直径大约在 400-700 码，180° 转向时间大约为 2min。

2min 对于 10min 的战术回合 20 节的速度，大约是 4 格（由于未考虑转向的航速损失等情况，此数值偏小），而 900 码的转向直径设置出于编队行动转向直径可能更大的考虑，因此也保留了单舰更小转向直径的选择。

不过这些对于尺规推演来说，想必会随着误差而不断累计。

[4]此文件内含有用于制作算子的俯视图，全部来源于公开互联网资料和电子书籍，压缩包内有每张图的引用来源和颜色的色号。

制作方式为收集俯视线图，ai 工具上色，归一化颜色。

[5]参考《AH Jutland rules thoughts and modifications (v5)》Brian McCue 其中 Tactical Movement 节对转向机动的论述。文件链接见上[2]。

[6]现实中由于转向的航速损失和海况等等情况，导致航速并不总能维持在最大航速。

[7]船表中主力舰的数据（HP、装甲、火炮等）的详情和来源可以打开文件夹 WorkBook 内 Jutland gun data.xlsx 和 Jutland ships.xlsx。

以下是其中记录的数据来源

《Die deutschen Kriegsschiffe 1815-1945》
《Battlecrusier》John Robert
《British Battleships of WW1》R A Burt
《The Battle of Jutland》John Brooks
https://www.bilibili.com/opus/1073001113983123475
https://www.bilibili.com/read/cv34684600
https://tieba.baidu.com/p/10710367839
https://www.warship1.cn/forum.php?mod=viewthread&tid=7352
https://www.warship1.cn/thread-7433-1-1.html
https://www.warship1.cn/thread-7435-1-1.html
https://tieba.baidu.com/p/10346601001
https://tieba.baidu.com/p/9289591876
https://tieba.baidu.com/p/8989921235
https://www.warship1.cn/thread-8542-1-4.html
https://www.bilibili.com/opus/1024120129801158724
https://tieba.baidu.com/p/7873575541
https://www.warship1.cn/thread-7423-1-1.html
http://www.navweaps.com/
https://www.warship1.cn/thread-6042-1-1.html
https://www.warship1.cn/thread-7495-1-1.html
https://www.warship1.cn/thread-7500-1-1.html
http://www.navweaps.com/index_inro/INRO_BB-Gunnery.php

[8]双方参战舰队的编成：
https://naval-history.net/WW1Battle-Battle_of_Jutland_1916_Official_Despatches1.htm

及《The Battle of Jutland》John Brooks

[9]战列舰主炮塔命名是否有标准？ - 米凯勒 法比恩的回答 - 知乎
https://www.zhihu.com/question/264368690/answer/2820124752

[10]参考 https://www.bilibili.com/opus/1073001113983123475
和 https://www.bilibili.com/read/cv34684600
以及 https://www.warship1.cn/forum.php?mod=viewthread&tid=7352
若以经验出发可以使用更低的航速数据，若追求准确可以使用更高的航速数据。
（笔者个人认为,如果采用更高航速可能出现英国战巡反而被德国战巡追逐的与历史区别的荒诞情况，建议将更高航速作为偶尔出现的情况）

[11]为了简化流程，此处将指挥塔装甲和炮塔装甲用同一个值表示。相关落弹角装甲数据，来源于简单的三角函数换算（显然不合理）和弹道估计，仍需进一步量化和理论分析。
详细的装甲汇编可见（Jutland ships.xlsx）
参考：https://www.warship1.cn/forum.php?mod=viewthread&tid=6153
https://tieba.baidu.com/p/9289591876

[12]来源：http://www.navweaps.com/
《British Battleships of WW1》R A Burt
《Battlecrusier》John Robert

[13]参考：《The Battle of Jutland》John Brooks

[14]来源：https://www.warship1.cn/thread-7518-1-1.html

[15]参考：《The Battle of Jutland》John Brooks
《British Destroyers From Earliest Days to the Second World War》Norman Friedman

[16]参考：http://www.navweaps.com/Weapons/WNBR_135-45_mk6.php
“1b.Superfiring turrets could not fire within 30 degrees of the axis because the blast effects would have penetrated into the lower turrets through the sighting hoods.”

又及 http://www.navweaps.com/Weapons/WNBR_12-50_mk11.php
“3d.The superfiring 'X' turret on these ships was fitted with stops to prevent it from firing over the top of 'Y' turret.”

[17]参考 http://www.navweaps.com/Weapons/WNBR_135-45_mk6.php
“7e."Q" turret on Tiger was in a more favorable position which increased its firing arc to 60 degrees before the beam and 90 degrees abaft it on either side, although it could not fire directly aft below about 10 degrees of elevation. Firing aft at angles between about 170 to 190 degrees would also affect wireless antennas on any British capital ship of this era.”

[18]参考 http://www.navweaps.com/Weapons/WNBR_12-50_mk11.php
“5d.HMS Neptune had significant "firsts" for British battleships. She was the first equipped with a superimposed turret, the first to be able to fire a broadside with all main guns and the first to be fitted with a director sight. None of these "firsts" were entirely successful. During initial gunnery trials it was found that when the amidships turrets fired cross-deck that the deck sagged appreciably from the blast effect. Extra pillars and "Z" bars were fitted to strengthen these areas. Astern and ahead fire was also restricted for all turrets to angles greater than 5 degrees in order to prevent blast damage to the deck and superstructure. ...”

又及 http://www.navweaps.com/Weapons/WNBR_12-45_mk10.php

“5b.HMS Dreadnought was designed so that the wing turrets could fire directly ahead or directly astern. However, it was found that the guns could not fire within 10 degrees of the axis without damaging the ship due to gun blast. The Invincible class had staggered amidships mountings intended to allow them to shoot across the ship as well as directly forward and directly aft. Gunnery trials showed that the blast damage from such firings was unacceptable and the actual firing arcs were restricted to approximately the figures given above. At the Falklands battle, Invincible fired cross-deck with both P and Q turret but this dazed and deafened the gun-layers, trainers and sight setters as well as causing blast damage to the deck. P turret reported that the trainers had to be replaced constantly during the battle as they were too dazed to function properly. After the battle, cross-deck firing was ruled out except in cases of emergency.”

[19]笔者对于大部分齐射角的测定来自于设计图线稿的测量，缺乏第一手的数据支持，若读者有更准确来源欢迎对此修正。

[20]参考 Grand Fleet Battle Orders

SECRET. 36
220—Dec. 15. Page 22.

GRAND FLEET BATTLE ORDERS.

Gunnery Instructions.

XV. DISTRIBUTION OF GUNFIRE (*continued*).

(b) Distribution of gunfire after a reversal of course.

7. After reversing our line of battle, or after reversal of the enemy line or if both lines are reversed, provided no change in the disposition of our squadrons has been made each of our divisions should continue to fire at the same numbered ships from the van or rear of the enemy as they were firing at before the turn, *i.e.*, the counting signal for the division is the same, but each individual ship of the division may have to shift her fire to a new enemy ship if the reversal of course has been made by a divisional turn in succession.

Example—

Both Fleets in line of battle as stated in the Example in paragraph 6.

Before turning.

5th Battle Squadron. ↑——↑ A, ↑——↑ B Enemy., ↑——↑ C

1st Division. Fleet No. 1 ↑——↑ D, ,, ,, 2 ↑——↑ E, ,, ,, 4 ↑——↑ F

After turning.

5th Battle Squadron. ↓——↓ D, ↓——↓ C, ↓——↓ B

1st Division. Fleet No. 4↓——↓ A, ,, ,, 2↓——↓ H, ,, ,, 1↓——↓ G

After turning, the Counting signal for the 1st Division is still the same, *i.e.*, L 4—6.
But Fleet No. 1 is firing at G., *i.e.*, the 6th ship from the left.
,, ,, ,, 2 ,, ,, ,, H., ,, 5th ,, ,, ,,
,, ,, ,, 4 ,, ,, ,, A., ,, 4th ,, ,, ,,

8. If, however, the enemy's battle-cruisers turn to take station in rear of their battlefleet followed by a turn of 16 points by the whole enemy fleet, and our 5th Battle Squadron follows the motions of the enemy's battle-cruisers as laid down in Section V, paragraph 12, page 11, then the disposition of our squadrons will be altered and a new counting signal will have to be made.

(c) Concentration.

9. When concentration is possible, *pairs* will give the best results, as mutual interference to observation of fire is then practically negligible; three or four ships can, however, concentrate effectively on one if the conditions are sufficiently good to admit of spotting and so keeping the range.

Ships such as those mentioned in paragraph 5 (*b*) of this section, which are not in the line of battle but are able to engage the enemy's battlefleet, should usually concentrate on the nearest enemy ships in preference to engaging singly a large number, some of which would be at a great range.

10. When an enemy ship hauls out of line, the ship which has been engaging her is to shift her fire ahead or astern according to where it seems to be most required. The points to be considered in selecting a new target are (*a*) where support is most needed, or (*b*) which enemy ship (in a suitable position) appears to be firing most effectively or to be little damaged. Changes in target due to enemy ships dropping out need not necessarily lead to ships other than the ship immediately concerned shifting their fire, at any rate until it is advantageous to do so.

11. If one of our ships hauls out of line, the ship astern must, if necessary, shift target so as not to leave any of the enemy's "*Dreadnoughts*" unfired at.

(d) Concentration on leading ships of columns.

12. If our fleet deploys before the German fleet, and on opening fire is in a position to concentrate on the leading ships of his columns, this should be done whether the signal '41' is made or not, *and until the enemy deploys and ships are able to engage their proper targets.*

13. In concentrating on the leading ships of columns, the angle of the enemy's approach may permit of the two ships at the head of a column being fired at, and if so, this should be done as a matter of course, one pair of a division of four ships taking the leading ship and the other the second ship.

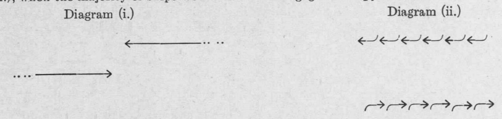
14. Each division must be primarily responsible for the enemy's column opposite in number to itself, but, if we have fewer divisions than the German fleet has columns, and all are within range, our centre divisions should take two columns each, leaving the van and rear divisions to deal with their opposite columns, which it is more important should suffer.

15. If we have more divisions than the Germans have columns, the fire of our divisions should be distributed so that the German columns composed of the most powerful ships receive the most fire.

(e) Distribution of gunfire when fleets meet and pass on opposite courses.

16. This condition, *vide* diagram (i.), which is referred to in the fourth clause of paragraph 2 of Home Fleets General Order No. 15, might occur in misty weather or at dawn, but it is not a likely one; it must not be confused with action on opposite courses which may follow an approach in columns on a broad front, *vide* diagram (ii.), when the majority of ships would be able to engage their opposite numbers from the outset).

Diagram (i.) Diagram (ii.)



In low visibility it is considered that only a very simple plan for distribution of gunfire is likely to answer, and it is better to leave as much as possible to the discretion of Captains.

The following is to be taken as a guide.—

"Each ship is to open fire in succession as the enemy comes into view, shifting her fire along the enemy's line shortly after a ship astern commences on the ship at which she is firing."

[17. However

[21]参考：https://tieba.baidu.com/p/9541816380
https://tieba.baidu.com/p/4250385818
https://www.warship1.cn/thread-6437-1-1.html
https://www.warship1.cn/thread-7499-1-1.html

笔者认为实战中射速的取值仍待商榷,若以 1/gun/min 的射速则海战中弹药消耗应当远超最终结果。相关原因推测可能有实战射击窗口有限、射界限制、交战距离远等。然而笔者也并没有获取第一手报告的资料。希望同好能就此做出批评指正并提出更好方案。

[22]参考 https://www.warship1.cn/thread-7499-1-1.html

[23]考虑此表对于正态分布拟合的误差偏大,玩家也可以采取投 D100 骰的办法来模拟。

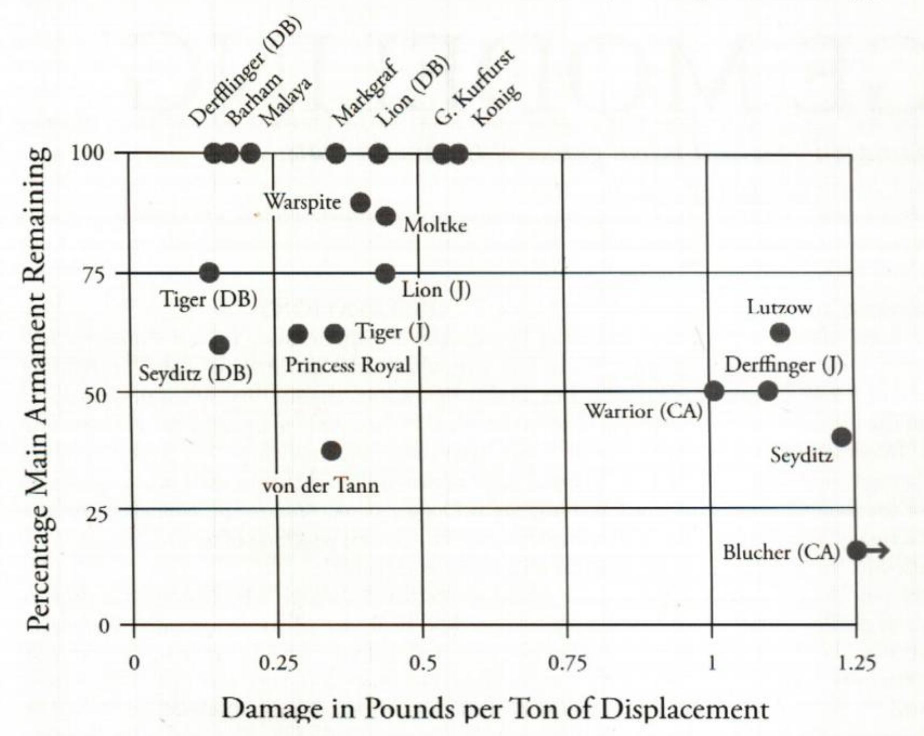
[24]此结算方式基于《败中取胜》即Moritsuchi 的《Solomon Night Naval Battle》（甲弹对抗）结算方式。

[25]基于 https://tieba.baidu.com/p/7873575541
即 https://www.warship1.cn/forum.php?mod=viewthread&tid=7587
中引用炮弹的爆炸威力，大致与其装药量的平方呈正比

然而考虑炮弹的装药和口径的立方呈正比，这一数据恐怕会增长到不甚合理的程度。

另一方面，《DAMAGE MODELING--An analysis of damage in tactical naval games of the world wars》John B. Gilmer Jr.中通过日德兰和多格尔沙洲数据的结果，认为船只承受损伤的能力相关于弹重。

FIGURE 1: DAMAGE, DOGGER BANK (DB) AND JUTLAND (J)



笔者在工作簿（Jutland gun data.xlsx）中就多种方式展开对比，最终选择以弹重、装药的正比来作为单发炮弹伤害的基准。

笔者认为在一战的炮弹引信的前提下（穿甲过程中早爆、穿甲后延时过短，参考战列舰论坛日德兰纪念系列），对船只的击沉除开发射药诱爆殉爆的情况，依赖于大量炮弹命中造成的伤害累计，而非一两发决定性命中。这一点应当有别于间战期及之后的情形。尽管一两

发决定性的命中往往能造成比大量无效命中更多的损伤,而且一战中纯粹依赖火炮的伤害累计击沉一艘大型船只案例也仅吕佐夫、布吕歇尔以及福克兰群岛两艘装甲巡洋舰。

对此问题的研究笔者自认为自己的观点不够严谨和可信,迫切希望具有高学术水平的同好、专家不吝惜正，就此进行详细批判或援引论证，提供可靠的解决方案。

[26]参考 https://www.warship1.cn/thread-7479-1-1.html

[27]参考《Destroyer Torpedo Attack Instruction》D.T.B. 4-43

[28]原作中并未予主力舰鱼雷点数,按照史实应当具备,用作主力舰近距离搏斗的手段。

[29]原始命中率参考自 http://www.navweaps.com/index_inro/INRO_BB-Gunnery.php，可见于（Jutland gun data.xlsx）中。

另外列出下列文献，在制作时阅读参考：

《Jutland —— an analysis of the fighting》John Campbell

《Naval Weapons of World War One》Norman Friedman

《The Grand Fleet: Warship Design and Development》D.K.Brown

其余还有战列舰吧(https://tieba.baidu.com/f?kw=%E6%88%98%E5%88%97%E8%88%B0)、战列舰论坛（https://www.warship1.cn/）等内容，不赘。

限于笔者学识与资料所及，疏漏谬误之处在所难免，

恳请各位同好、读者不吝指正！